

DTSA 5513

Deep Learning for Computer Vision

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (70.3 / 100)	60.0 / 100
SPECIALIZATION Computer Vision		

60	96	+45
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

This course introduces students to the core principles of neural networks and deep learning, focusing on their application in computer vision. Covering advanced CNN architectures like ResNet, Inception, and DenseNet, along with techniques in object detection (R-CNN, SSD, YOLO) and semantic segmentation (FCN, SegNet, U-Net), this course offers a comprehensive overview of theory and practical skills.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	70.3	100.0	+29.7	MEDIUM
AI Engineer	62.5	96.7	+34.3	HIGH
ML Engineer	62.1	93.4	+31.3	HIGH
Applied Scientist	57.4	100.0	+42.6	HIGH
MLOps Engineer	55.5	91.0	+35.5	HIGH
Data Scientist	52.4	97.6	+45.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Experiment Tracking 1 of 6 roles

tracking

Deep Learning 1 of 6 roles

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 LOW Optional: Experiment Tracking — would broaden appeal to one additional role. Candidate terms: tracking.

3 LOW Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: training.

4 LOW Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

70.3 → 100.0 (+29.7 pts)

data sets analytics tools business mobile design science

AI Engineer

62.5 → 96.7 (+34.3 pts)

data models build technical systems problems technology machine

ML Engineer

62.1 → 93.4 (+31.3 pts)

discovery data machine systems models world build problems

Applied Scientist

57.4 → 100.0 (+42.6 pts)

discovery scientific research literature diffusion problems models technical

MLOps Engineer

55.5 → 91.0 (+35.5 pts)

data text systems machine tracking image training tools

Data Scientist

52.4 → 97.6 (+45.2 pts)

data sets analytics different social build problems approaches

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.